

Program: Diploma in Food Processing Technology	
Course Code: 5428	Course Title: Food additives Lab
Semester: 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3(L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Determine the effects of various samples additives in the sample
- Study the presence or estimation of various additives in sample
- Extraction of additives from natural sources

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Awareness about food additive		Food additive	5

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Apply estimation techniques to assess the effects of preservatives, nonnutritive sweeteners, antioxidants, thickeners, anticaking agents, and acidulants in foods.	17	Applying
CO2	Estimate color in food samples	5	Applying
CO3	Determine the presence of adulterants in sample	9	Applying
CO4	Extract food additives from natural sources	14	Applying

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Blooms Taxonomy Level
CO1	Apply estimation techniques to assess the effects of preservatives, nonnutritive sweeteners, antioxidants, thickeners, anticaking agents, and acidulants in foods.		
M1.01	Estimation of preservatives and their effects in foods.	2	Understanding
M1.02	Estimation of nonnutritive sweeteners and their effect in foods	3	Applying
M1.03	Estimation of effects of antioxidants usage in foods.	3	Applying
M1.04	Estimation of Thickeners and their effects in foods.	3	Applying
M1.05	Observation of the effect of anticaking agents in foods	3	Applying
M1.06	Study of Acidulants in foods and their effects.	3	Applying
CO2	Estimate color in food samples		
M2.01	Estimation of Colors in foods.	3	Applying
	Lab Test –I	2	
CO3	Determine the presence of adulterants in sample		

M3.01	Detection/Estimation of adulterants in some foods	3	Applying
M3.02	Visit to a food processing industry	6	Applying
CO4	Extract food additives from natural sources		
M4.01	Extraction of essential oils	3	Applying
M4.02	Extraction of color from natural sources	6	Applying
M4.03	Extraction of flavorings from natural sources	3	Applying
	Lab Test– II	2	

Text/Reference

T/R	Book Title/Author
R1	ISI hand book of food analysis
R2	Hand book of analysis and quality control for fruit and vegetable products, by S.Ranganna, II Ed., Tata McGraw Hill Publishing Co. New Delhi.
R3	Official Method of analysis of AOAC

Online Resources

Sl.No	Website Link
1.	https://www.allianceingredients.in/products/acidulants-as-food-additives/28
2.	https://www.researchgate.net/publication/329058662_Extraction_of_Natural_Dyes_from_Plants
3.	https://www.scienceofcooking.com/food-thickening-agents.html